In [1]:
```python
import pandas as pd
from IPython.display import display
df = pd.read_csv("D:\\Data Engineering\\SQL\\Ele_Store1.csv")
```

In [3]:
```python
df.head(1)
```

Out[3]:

	OrderID	OrderDate	UnitCost	Price	OrderQty	CostofSales	Sales	Profit	Chan
0	7077	13-09-2017	76.094968	304.0	9	684.85471	2714.72	2029.86529	St

In [ ]:
```python
#1) Query the list of columns?
```

In [6]:
```python
list([df.columns])
```

Out[6]:
```
[[Index(['OrderID', 'OrderDate', 'UnitCost', 'Price', 'OrderQty', 'Costo
fSales',
         'Sales', 'Profit', 'Channel', 'PromotionName', 'ProductName',
         'Manufacturer', 'ProductSubCategory', 'ProductCategory', 'Regio
n',
         'City', 'Country'],
        dtype='object')]]
```

In [ ]:
```python
#2) Query the top 5 rows?
```

In [7]:
```
df.head(5)
```

Out[7]:

	OrderID	OrderDate	UnitCost	Price	OrderQty	CostofSales	Sales	Profit
0	7077	13-09-2017	76.094968	304.00	9	684.854710	2714.7200	2029.865290
1	117	14-09-2017	7.491753	12.99	4	29.967011	50.1414	20.174389
2	7018	15-09-2017	10.122338	159.99	9	91.101039	1395.1128	1304.011761
3	140	16-09-2017	0.576153	25.69	18	10.370759	462.4200	452.049241
4	491	17-09-2017	108.508777	304.00	9	976.578991	2614.4000	1637.821009

In [ ]:
```
#3) Query the bottom 5 rows?
```

In [8]:
```
df.tail(5)
```

Out[8]:

	OrderID	OrderDate	UnitCost	Price	OrderQty	CostofSales	Sales	Profit
14995	22348	03-10-2058	16.240455	25.0	12	194.885455	300.0	105.114545
14996	22349	04-10-2058	55.387212	999.0	10	553.872121	9990.0	9436.127879
14997	22350	05-10-2058	363.634986	588.0	13	4727.254816	7408.8	2681.545184
14998	22351	06-10-2058	113.237461	279.0	12	1358.849536	3348.0	1989.150464
14999	22352	07-10-2058	10.936943	15.9	20	218.738862	318.0	99.261138

In [ ]:
```
#4) What is total number of orders placed?
```

In [71]:
```python
orderscount = df[["OrderID"]].count()
print("Total number of orders :",orderscount)
```

```
'Total number of orders :'

OrderID    15000
dtype: int64
```

In [ ]:
```python
#5) What are the unique channels present and how many are there?
```

In [23]:
```python
uniquechannels = df["Channel"].unique()
print("The distinct channels are :",uniquechannels)
uniquecount = df["Channel"].nunique()
print("The count of distinct channels are:",uniquecount)
```

```
The distinct channels are : ['Store' 'Online' 'Reseller' 'Catalog']
The count of distinct channels are: 4
```

In [25]:
```python
#6) How many cities are there?
```

In [28]:
```python
citycount = df["City"].nunique()
print("Total cities :",citycount)
```

```
Total cities : 263
```

In [ ]:
```python
#7) Query the unique Countries and count of the countries?
```

In [30]:
```python
uniquecountry = df["Country"].unique()
print("The countries available are :",uniquecountry)
countrycount = df["Country"].nunique()
print("The total count of countries are: ",countrycount)
```

```
The countries available are : ['Russia' 'United States' 'China' 'United
Kingdom' 'Canada' 'Germany'
 'Turkmenistan' 'France' 'Australia' 'Iran' 'Thailand' 'Kyrgyzstan'
 'South Korea' 'Pakistan' 'India' 'Japan' 'Ireland' 'Singapore' 'Armeni
a'
 'Greece' 'Romania' 'Denmark' 'Italy' 'Malta' 'Portugal' 'Slovenia'
 'Syria' 'Switzerland' 'Taiwan' 'the Netherlands' 'Poland' 'Bhutan'
 'Spain' 'Sweden']
The total count of countries is:  34
```

In [ ]:
```python
#8) Query the unique Regions and count of the regions?
```

In [31]:
```
regionnames = df["Region"].unique()
print("The Regions available are :",regionnames)
regioncount = df["Region"].nunique()
print("The total count of regions are :",regioncount)
```

```
The Regions available are : ['Europe' 'North America' 'Asia']
The total count of regions are : 3
```

In [ ]:
```
#9) Query the total cities count by Country and order it from highest coun
```

In [33]:
```
df.groupby(["Country"])[["City"]].count().sort_values("City",ascending = F
```

Out[33]:

Country	City
United States	8510
China	1673
Germany	1118
France	781
United Kingdom	477
Canada	359
Japan	298
Russia	166
India	152
Australia	149
Turkmenistan	119
Italy	111
Pakistan	93
Iran	89
Syria	86
Thailand	82
South Korea	72
Taiwan	50
Singapore	49
Bhutan	48
Kyrgyzstan	46
Armenia	44
Denmark	42
Malta	41
Romania	39
the Netherlands	38
Portugal	38
Ireland	37
Poland	37
Switzerland	35
Slovenia	34
Spain	32
Sweden	30
Greece	25

In [ ]:
```
#10) How many products are there? #Display count of products
```

In [35]:
```
productcount = df["ProductName"].nunique()
print("The count of product are :",productcount )
```

```
The count of product are : 1638
```

In [ ]:
```
#11) Query the total sales and total profit?
```

In [70]:
```
salesProfit = df[["Sales","Profit"]].sum()
print("The total sales is :", salesProfit[0])
print("The total prfoit is :", salesProfit[1] )
```

```
'The total sales is :'

55391759.788100004

The total prfoit is : 31587437.287211657
```

In [ ]:
```
#12) Query the total sales by region and country?
```

In [69]:
```python
RegncntrySales=df.groupby(["Region","Country"])[["Sales"]].sum().sort_valu
display(RegncntrySales)
```

Region	Country	Sales
North America	United States	3.163524e+07
Asia	China	7.557974e+06
Europe	Germany	4.314956e+06
	France	2.628401e+06
	United Kingdom	1.324267e+06
North America	Canada	1.148316e+06
Asia	Japan	9.583344e+05
	Australia	6.326425e+05
	India	5.984046e+05
Europe	Russia	4.342403e+05
Asia	Turkmenistan	4.123702e+05
	Iran	3.679533e+05
	Syria	3.296714e+05
Europe	Italy	3.168811e+05
Asia	Pakistan	2.965706e+05
	Thailand	2.582489e+05
	South Korea	2.506013e+05
	Taiwan	1.890095e+05
	Armenia	1.729347e+05
	Bhutan	1.612916e+05
Europe	Spain	1.434332e+05
Asia	Kyrgyzstan	1.271148e+05
Europe	Portugal	1.263816e+05
	Malta	1.122952e+05
	the Netherlands	1.077465e+05
	Romania	1.022789e+05
	Switzerland	9.936224e+04
Asia	Singapore	9.922349e+04
Europe	Ireland	9.212368e+04
	Poland	8.782953e+04
	Slovenia	7.988141e+04
	Denmark	7.914523e+04
	Sweden	7.810665e+04
	Greece	6.852796e+04

In [ ]:
```
#13) Query the total order qty by promotion name?
```

In [68]:
```
promotionordrqty = df.groupby(["PromotionName"])[["OrderQty"]].sum()
display(promotionordrqty)
#display command to shoe the result in tabular format.for this we need to
```

	OrderQty
PromotionName	
Asian Holiday Promotion	17004
Asian Spring Promotion	10980
Asian Summer Promotion	9768
European Back-to-Scholl Promotion	7363
European Holiday Promotion	12328
European Spring Promotion	13205
No Discount	80062
North America Back-to-School Promotion	39393
North America Holiday Promotion	30392
North America Spring Promotion	30631

In [ ]:
```
#14) query the top 7 countries with sales from highest to lowest order?
```

In [74]:
```
df.groupby(["Country"])[["Sales"]].sum().sort_values("Sales",ascending = F
```

Out[74]:

	Sales
Country	
United States	3.163524e+07
China	7.557974e+06
Germany	4.314956e+06
France	2.628401e+06
United Kingdom	1.324267e+06
Canada	1.148316e+06
Japan	9.583344e+05

In [77]:
```
df.groupby(["Country"])[["Sales"]].sum().sort_values("Sales",ascending = F
```

Out[77]:

Country	Sales
United States	3.163524e+07
China	7.557974e+06
Germany	4.314956e+06
France	2.628401e+06
United Kingdom	1.324267e+06
Canada	1.148316e+06
Japan	9.583344e+05

In [ ]:
```
#15) How much the total sales happened in Asia region?
```

In [7]:
```
asiasales = df[df["Region"] == "Asia"]["Sales"].sum()
print("The Asia regions sales :",asiasales)
```

```
The Asia regions sales : 12412345.056499999
```

In [ ]:
```
#16) How much total sales happend in Asia region and India?
```

In [13]:
```
asiaindiasales = df.query('Region == "Asia" & Country == "India"')["Sales"
print("The total sales happend in Asia and India :",asiaindiasales )
```

```
The total sales happend in Asia and India : 598404.6095
```

In [4]:
```
#17) How much profit got in Eurpoe region and Denmark country?
```

In [7]:
```
europedenmarkprofit = df.query('Region == "Europe" & Country== "Denmark"')
print("The total profit in Europe and Denmark :",europedenmarkprofit )
```

```
The total profit in Europe and Denmark : 79145.2295
```

In [14]:
```
eudenprofit = df.loc[(df["Region"] == "Europe") & (df["Country"] == "Denma
print("The total profit in Europe and Denmark :",eudenprofit)
```

```
The total profit in Europe and Denmark : 79145.2295
```

In [ ]:
```
#18) Query the total order qty by Manufacturer and Product Name?
```

In [23]:
```
manprodaordrqty = df.groupby(["Manufacturer","ProductName"])[["OrderQty"]]
display(manprodaordrqty)
```

Manufacturer	ProductName	OrderQty
A. Datum Corporation	A. Datum Advanced Digital Camera M300 Azure	77
	A. Datum Advanced Digital Camera M300 Black	142
	A. Datum Advanced Digital Camera M300 Green	96
	A. Datum Advanced Digital Camera M300 Grey	207
	A. Datum Advanced Digital Camera M300 Orange	115
...	...	...
Wide World Importers	WWI Wireless Transmitter and Bluetooth Headphones X250 Black	45
	WWI Wireless Transmitter and Bluetooth Headphones X250 Blue	65
	WWI Wireless Transmitter and Bluetooth Headphones X250 Red	60
	WWI Wireless Transmitter and Bluetooth Headphones X250 Silver	60
	WWI Wireless Transmitter and Bluetooth Headphones X250 White	48

1638 rows × 1 columns

In [ ]:
```
#19) Query the top 7 products with orderqty and sales and show it descendi
```

In [30]:
```
df.groupby(["ProductName"])[["OrderQty","Sales"]].sum().sort_values("Order
```

Out[30]:

ProductName	OrderQty	Sales
Contoso In-Line Coupler E180 Silver	6780	22696.2500
Headphone Adapter for Contoso Phone E130 Silver	4680	46730.9223
Contoso Rubberized Snap-On Cover Hard Case Cell Phone Protector E160 Silver	4580	21700.9050
Contoso In-Line Coupler E180 White	4060	13588.4375
Contoso Rubberized Skin BlackBerry E100 Silver	3560	53319.8797
Cigarette Lighter Adapter for Contoso Phones E110 White	3520	87912.3210
Contoso Rubberized Snap-On Cover Hard Case Cell Phone Protector E160 Pink	3280	15534.4020

In [ ]:
```
#20) Query the total order quantity greater than 10000 by city?
```

In [49]:
```
cityorderqty = df.groupby(["City"])[["OrderQty"]].sum().sort_values("Order
display(cityorderqty)
```

	OrderQty
City	
Beijing	30547
North Harford	24806
Bethesda	20498
Berlin	13841
Seattle	12174

In [ ]:
```
#21) Query the total sales less than 1000 by product, display it in ascend
```

In [55]:

```
df.groupby(['ProductName'])[["Sales"]].sum().sort_values("Sales",ascending
```

Out[55]:

ProductName	Sales
SV USB Data Cable E600 Grey	45.6950
SV USB Sync Charge Cable E700 Silver	64.0780
SV USB Data Cable E600 Pink	100.4340
SV USB Data Cable E600 Black	124.2600
SV USB Sync Charge Cable E700 Blue	142.6830
SV USB Sync Charge Cable E700 Black	160.3940
SV USB Sync Charge Cable E700 White	170.7420
SV USB Data Cable E600 Silver	202.8060
Contoso Primary Extended Capacity Battery Pack - notebook battery X100 White	203.4000
Contoso USB Wave Multi-Media Keyboard M901 Gold	259.0000
Contoso Digital Cordless Expansion Handset Phone M900 Grey	293.6110
Contoso Wireless Notebook Optical Mouse M35 White	299.5000
Contoso Digital Cordless Expansion Handset Phone M900 White	329.9000
Contoso Reserve Pen - Tablet Pen E200 Black	332.2100
Contoso Ultraportable Neoprene Sleeve E30 Green	335.4000
Contoso Phone with Memory Dialing-2 lines E90 Grey	382.4000
Contoso Wireless Notebook Optical Mouse M35 Orange	383.3600
Contoso 4-Line Corded Cordless Telephone M203 Grey	395.8800
Contoso Lens Cap Keeper E314 Silver	415.6100
Contoso DVD Player M100 Black	504.3615
Contoso General Carrying Case E304 White	511.7440
Contoso Digital Cameras Lightweight Tripod E316 Silver	521.3000
Contoso Digital Camera Accessory kit M200 Blue	525.8000
Contoso Education Supplies Bundle E200 Black	532.5750
Contoso Smart Battery M901 Blue	547.4000
Contoso Lens cap E80 White	599.0000
Contoso DVD Player M140 Gold	609.9000
Contoso USB Optical Mouse E200 Gold	669.6000
Contoso USB Wave Multi-media Keyboard E280 Blue	670.9800
Contoso Smart Battery M901 Black	672.1750
Contoso Reserve Pen - Tablet Pen E200 White	678.3200
Contoso ADSL Modem Splitter/Filter X 1 E100 White	680.2000
SV DVD Player M130 Grey	713.8810
SV DVD 58 DVD Storage Binder M55 Silver	738.9480
Contoso Lens Cap Keeper E314 White	749.2795
Contoso Wireless Notebook Optical Mouse X205 Orange	766.7050
Contoso Lens cap E80 Black	778.7000
SV DVD 38 DVD Storage Binder E25 Silver	816.1830

ProductName	Sales
Contoso Laptop Starter Bundle M200 White	837.0450
Contoso General Soft Carrying Case E318 Silver	867.1715
Contoso Laptop Starter Bundle M200 Grey	891.0000
Contoso Laptop Starter Bundle M200 Black	898.4250
Contoso Cyber Shot Digital Cameras Adapter E306 Pink	901.7745
Contoso Private Branch Exchange M88 White	912.1735
Contoso Desktop Alternative Bundle E200 Grey	915.4000
Contoso General Soft Carrying Case E318 Blue	929.3800
Contoso Wireless Notebook Optical Mouse M35 Silver	949.4150
Contoso Phone with Memory Dialing-2 lines E90 Black	960.0000
Contoso Education Essentials Bundle M300 White	994.2700
Contoso Reserve Pen - Tablet Pen E200 Grey	996.6300

In [ ]:
```
#22) Query the total profit greater than 200000 by product, display it in
```

In [58]:
```
df.groupby(["ProductName"])[["Profit"]].sum().sort_values("Profit",ascendi
```

Out[58]:

ProductName	Profit
Contoso Projector 1080p X980 White	320513.293550
Proseware Projector 1080p LCD86 White	315379.915450
Contoso Projector 1080p X980 Black	252654.597032
Proseware Projector 1080p LCD86 Black	232556.514538
Proseware Projector 1080p DLP86 Silver	228823.614999
Proseware Projector 1080p DLP86 Black	208705.101207

In [ ]:
```
#23) How much total sales happened in China and Beijing?
```

In [63]:
```
chbjsales = df.query('Country == "China" & City == "Beijing"')["Sales"].su
print("The total sales happend in china and beijing is :",chbjsales )
```

```
The total sales happend in china and beijing is : 6596953.0955
```

In [ ]:
```
#24) How much total sales happened in Asian Holiday Promotion?
```

In [65]:
```
ashopromoslaes = df.query('PromotionName == "Asian Holiday Promotion"')["S
print("The total sales happend in Asian Holiday Promotion is :",ashopromos
```

```
The total sales happend in Asian Holiday Promotion is : 3615193.99050000
03
```

In [ ]:
```
#25) How much total profit got from Contoso, Ltd Manufacturer?
```

In [71]:
```
ContosoLtdprofit = df.query('Manufacturer == "Contoso, Ltd"')["Profit"].su
print("The Contoso Ltd manufacturer profit is :",ContosoLtdprofit)
```

```
The Contoso Ltd manufacturer profit is : 5670159.660158278
```

In [ ]:
```
#26) Query the total sales and total order qty by productcategory and orde
```

In [85]:
```
df.groupby(["ProductCategory"])[["Sales","OrderQty"]].sum()
```

Out[85]:

ProductCategory	Sales	OrderQty
Audio	1.029381e+06	9067
Cameras and camcorders	1.704893e+07	43893
Cell phones	5.841156e+06	96074
Computers	2.130697e+07	69425
Music, Movies and Audio Books	1.057480e+06	9953
TV and Video	9.107839e+06	22714

In [ ]:
```
#27) Query the total sales value in between 1057 and 26700 by Region?
```

In [87]:
```
df[["Region","Sales"]].query('Sales > 1057 and Sales < 26700')
```

Out[87]:

	Region	Sales
0	Europe	2714.7200
2	Europe	1395.1128
4	Asia	2614.4000
5	Asia	7056.4000
6	Asia	6990.0000
...	...	...
14991	North America	3380.0000
14992	North America	3510.0000
14996	Europe	9990.0000
14997	North America	7408.8000
14998	Asia	3348.0000

10130 rows × 2 columns

In [ ]:
```
#28) How many orders placed in Pittsfield city?
```

In [92]:
```
Pittsfieldcityorders =  df.loc[df["City"] == "Pittsfield"]["OrderID"].cour
print("The total orders placed in Pittsfield city is :",Pittsfieldcityorde
```

```
The total orders placed in Pittsfield city is : 31
```

In [ ]:
```
#29) Query the total sales and total profit by region, country and city?
```

In [93]:
```
df.groupby(["Region","Country","City"])[["Sales","Profit"]].sum()
```

Out[93]:

Region	Country	City	Sales	Profit
Asia	Armenia	Yerevan	1.729347e+05	1.119702e+05
	Australia	Canberra	2.112396e+05	9.384047e+04
		Sydney	4.214029e+05	2.503940e+05
	Bhutan	Thimphu	1.612916e+05	8.817516e+04
	China	Beijing	6.596953e+06	3.810007e+06
...	...	...	...	...
North America	United States	Waukesha	1.953965e+05	9.556164e+04
		Wheat Ridge	1.295698e+05	5.794578e+04
		Winchester	9.426428e+04	5.261520e+04
		Worcester	1.839504e+05	1.087016e+05
		Yakima	1.008435e+05	7.079848e+04

263 rows × 2 columns

In [ ]:
```
#30) Query the total unit cost and total price by productcategory, product
```

In [95]:

```
df.groupby(["ProductCategory","ProductSubCategory","Manufacturer"])[["Unit
```

Out[95]:

ProductCategory	ProductSubCategory	Manufacturer	UnitCost	Price
Audio	Bluetooth Headphones	Northwind Traders	4694.140924	10698.36
		Wide World Importers	5348.553816	12740.88
	MP4&MP3	Contoso, Ltd	15180.676953	37464.56
	Recording Pen	Wide World Importers	10165.599158	22585.30
Cameras and camcorders	Camcorders	Fabrikam, Inc.	286756.395516	705377.00
	Cameras & Camcorders Accessories	Contoso, Ltd	12221.056068	26301.12
	Digital Cameras	A. Datum Corporation	88363.414849	210914.00
	Digital SLR Cameras	A. Datum Corporation	55606.839775	148662.00
		Contoso, Ltd	61069.829295	155672.00
		Fabrikam, Inc.	64966.210095	174022.20
Cell phones	Cell phones Accessories	Contoso, Ltd	1526.510851	3530.08
	Home & Office Phones	Contoso, Ltd	12044.658547	27389.09
	Smart phones & PDAs	The Phone Company	103969.796452	239147.00
	Touch Screen Phones	Contoso, Ltd	12851.720868	27104.00
		The Phone Company	55433.985480	130579.00

ProductCategory	ProductSubCategory	Manufacturer	UnitCost	Price
Computers	Computers Accessories	Contoso, Ltd	16153.021031	35657.80
		Southridge Video	12789.845684	24305.89
	Desktops	Adventure Works	43576.939570	89568.35
		Wide World Importers	31821.889351	64659.00
	Laptops	Adventure Works	93609.664408	224875.55
		Fabrikam, Inc.	52102.113425	127405.35
		Proseware, Inc.	26831.171937	59963.95
		Wide World Importers	45528.259167	119908.70
	Monitors	Adventure Works	23132.559566	54600.00
		Proseware, Inc.	17530.704507	42299.00
		Wide World Importers	22386.928271	58944.00
	Printers, Scanners & Fax	Proseware, Inc.	65123.253204	155502.00
	Projectors & Screens	Contoso, Ltd	78241.936027	198399.00
		Proseware, Inc.	99812.740227	259541.00
		Wide World Importers	62842.407007	146677.00
Music, Movies and Audio Books	Movie DVD	Contoso, Ltd	16690.866047	44897.45
		Southridge Video	17559.035965	42566.69
TV and Video	Car Video	Southridge Video	84685.148227	179049.00
	Home Theater System	Contoso, Ltd	97944.784475	205953.00
		Litware, Inc.	87195.387626	187306.80
	Televisions	Adventure Works	71821.432011	174949.73
	VCD & DVD	Southridge Video	8797.508772	18504.60

In [99]:
```
df.shape
```

Out[99]: (15000, 17)